

Android Studio

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Code 1: Android Fragment

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical">

    <Button
        android:id="@+id/btnFirst"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Show First Fragment" />

    <Button
        android:id="@+id/btnSecond"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Show Second Fragment" />

    <FrameLayout
        android:id="@+id/fragmentContainer"
        android:layout_width="match_parent"
        android:layout_height="match_parent" />

</LinearLayout>
```

```
package com.example.project1java;

import android.os.Bundle;
import android.view.View;

import androidx.appcompat.app.AppCompatActivity;
import androidx.fragment.app.Fragment;
import androidx.fragment.app.FragmentManager;
import androidx.fragment.app.FragmentTransaction;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);

        setContentView(R.layout.activity_main);

        findViewById(R.id.btnFirst).setOnClickListener(new
View.OnClickListener() {
    @Override
```

```

        public void onClick(View v) {

            loadFragment(new FirstFragment());

        }
    });

    findViewById(R.id.btnSecond).setOnClickListener(new
View.OnClickListener() {
    @Override
    public void onClick(View v) {

        loadFragment(new SecondFragment());

    }
});
}

private void loadFragment(Fragment fragment) {

    FragmentManager fragmentManager = getSupportFragmentManager();

    FragmentTransaction transaction =
        fragmentManager.beginTransaction();

    transaction.replace(R.id.fragmentContainer, fragment);

    transaction.commit();

}
}

```

```

package com.example.project1java;

import android.os.Bundle;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;

import androidx.fragment.app.Fragment;

public class FirstFragment extends Fragment {

    @Override
    public View onCreateView(LayoutInflater inflater,
                             ViewGroup container,
                             Bundle savedInstanceState) {

        View view = inflater.inflate(R.layout.first_fragment, container,
false);

        return view;

    }
}

```

```

package com.example.project1java;

import android.os.Bundle;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;

import androidx.fragment.app.Fragment;

public class SecondFragment extends Fragment {

    @Override
    public View onCreateView(LayoutInflater inflater,
                             ViewGroup container,
                             Bundle savedInstanceState) {

        View view = inflater.inflate(R.layout.second_fragment, container,
false);

        return view;
    }
}

```

```

<?xml version="1.0" encoding="utf-8"?>

<FrameLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_gravity="center"
        android:text="This is First Fragment"
        android:textSize="20sp"
        android:textColor="#1976D2" />

</FrameLayout>

```

```

<?xml version="1.0" encoding="utf-8"?>

<FrameLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_gravity="center"
        android:text="This is Second Fragment"
        android:textSize="20sp"
        android:textColor="#D32F2F" />

</FrameLayout>

```

```
</FrameLayout>
```

Code 2:CardView

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">

    <!-- First Card -->
    <androidx.cardview.widget.CardView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_margin="16dp"
        app:cardCornerRadius="12dp"
        app:cardElevation="4dp">

        <LinearLayout
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:orientation="horizontal"
            android:padding="16dp">

            <ImageView
                android:layout_width="48dp"
                android:layout_height="48dp"
                android:src="@drawable/ic_person"
                android:tint="@android:color/holo_blue_dark" />

            <LinearLayout
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:layout_marginStart="16dp"
                android:orientation="vertical">

                <TextView
                    android:layout_width="wrap_content"
                    android:layout_height="wrap_content"
                    android:text="Student"
                    android:textSize="18sp"
                    android:textStyle="bold" />

                <TextView
                    android:layout_width="wrap_content"
                    android:layout_height="wrap_content"
                    android:text="View your profile"
                    android:textSize="14sp" />

            </LinearLayout>

        </LinearLayout>

    </androidx.cardview.widget.CardView>

</LinearLayout>
```

```

        </LinearLayout>

    </LinearLayout>

</androidx.cardview.widget.CardView>

<!-- Second Card -->
<androidx.cardview.widget.CardView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_margin="16dp"
    app:cardCornerRadius="12dp"
    app:cardElevation="4dp">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="horizontal"
        android:padding="16dp">

        <ImageView
            android:layout_width="48dp"
            android:layout_height="48dp"
            android:src="@drawable/ic_book"
            android:tint="@android:color/holo_blue_dark" />

        <LinearLayout
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_marginStart="16dp"
            android:orientation="vertical">

            <TextView
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:text="Courses"
                android:textSize="18sp"
                android:textStyle="bold" />

            <TextView
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:text="Explore available courses"
                android:textSize="14sp" />

        </LinearLayout>

    </LinearLayout>

</androidx.cardview.widget.CardView>

<!-- Third Card -->
<androidx.cardview.widget.CardView
    android:layout_width="match_parent"

```

```

        android:layout_height="wrap_content"
        android:layout_margin="16dp"
        app:cardCornerRadius="12dp"
        app:cardElevation="4dp">

        <LinearLayout
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:orientation="horizontal"
            android:padding="16dp">

            <ImageView
                android:layout_width="48dp"
                android:layout_height="48dp"
                android:src="@drawable/ic_setting"
                android:tint="@android:color/holo_blue_dark" />

            <LinearLayout
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:layout_marginStart="16dp"
                android:orientation="vertical">

                <TextView
                    android:layout_width="wrap_content"
                    android:layout_height="wrap_content"
                    android:text="Settings"
                    android:textSize="18sp"
                    android:textStyle="bold" />

                <TextView
                    android:layout_width="wrap_content"
                    android:layout_height="wrap_content"
                    android:text="Manage app preferences"
                    android:textSize="14sp" />

            </LinearLayout>

        </LinearLayout>

    </androidx.cardview.widget.CardView>

</LinearLayout>

```

```

package com.example.project1java;

import android.os.Bundle;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
    }
}

```

```

        setContentView(R.layout.activity_main);
    }
}

```

Code 3:Progress bar

```

<?xml version="1.0" encoding="utf-8"?>

<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="16dp">

    <ProgressBar
        android:id="@+id/progressBar"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:visibility="gone" />

    <Button
        android:id="@+id/btnStart"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Start Task" />

</LinearLayout>

```

```

package com.example.project1java;

import android.os.Bundle;
import android.os.Handler;
import android.view.View;
import android.widget.Button;
import android.widget.ProgressBar;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    ProgressBar progressBar;
    Button btnStart;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);

        setContentView(R.layout.activity_main);

        progressBar = findViewById(R.id.progressBar);
        btnStart = findViewById(R.id.btnStart);
    }
}

```

```

        btnStart.setOnClickListener(new View.OnClickListener() {

            @Override
            public void onClick(View v) {

                progressBar.setVisibility(View.VISIBLE);

                new Handler().postDelayed(new Runnable() {

                    @Override
                    public void run() {

                        progressBar.setVisibility(View.GONE);

                    }

                }, 3000);

            }

        });
    }
}

```

Code 4: Themes

```

<?xml version="1.0" encoding="utf-8"?>

<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/mainLayout"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="24dp">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Hello Material Design!"
        android:textSize="24sp"
        android:textStyle="bold" />

    <com.google.android.material.button.MaterialButton
        android:id="@+id/btnClick"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="16dp"
        android:text="Click Me" />

</LinearLayout>

```

```
package com.example.project1java;

import android.os.Bundle;
import android.graphics.Color;
import android.view.View;

import androidx.appcompat.app.AppCompatActivity;
import com.google.android.material.button.MaterialButton;

public class MainActivity extends AppCompatActivity {

    View mainLayout;
    MaterialButton btnClick;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);

        setContentView(R.layout.activity_main);

        mainLayout = findViewById(R.id.mainLayout);
        btnClick = findViewById(R.id.btnClick);

        btnClick.setOnClickListener(new View.OnClickListener() {

            @Override
            public void onClick(View v) {

                mainLayout.setBackgroundColor(Color.BLACK);

            }
        });
    }
}
```